

Lily Shen

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EDUCATION

University of Maryland, College Park

Expected May 2028

Bachelor of Science, Computer Science | Minor in General Business

GPA: 3.89/4.0

- **Advanced Cybersecurity Experience for Students (ACES) Honors College:** ~70 students selected university-wide.
- **Banneker/Key Scholar - Full-ride merit scholarship** awarded to top 0.1% of undergraduate students at UMD.
- **Relevant Coursework:** Algorithms, Computer Systems, Object-Oriented Programming I & II, Programming Languages, Discrete Structures, Applied Cybersecurity

EXPERIENCE

General Dynamics Information Technology

Jun 2026 - Present

AI/ML Engineering Intern

Springfield, VA

- Architect and deploy a PyTorch library for 3D LiDAR semantic segmentation across **13 classes** and PLY/LAS/LAZ formats.
- Optimize LiDAR scan inference with spatial tiling, **10-pass** voting, and reversible voxelization for **70M-point** scans.
- Validate inference on **10–60M-point** Leica LiDAR scans using a pretrained S3DIS model achieving ~**63%** mIoU.

Exiger (Client Project)

Feb 2026 - May 2026

Full-Stack Engineering Intern

College Park, MD

- Engineered **10+ reusable React components** for an enterprise supplier portal serving **1M+** suppliers.
- Built data tables and messaging functionality enabling downstream customer visibility across component supply chains.

Replenish NutrAI

Dec 2025 - Jun 2026

Backend Engineering Intern

Remote

- Designed a **7-layer** automated backend validation pipeline to audit active ingredient formulations across **850+ commercial products**, cutting manual verification overhead by **60%**.
- Implemented backend schemas integrating **650+** JSON-formatted clinical trials and synonym-normalized compound datasets.

Johns Hopkins Applied Physics Laboratory (APL)

Sep 2023 - Aug 2024

Full-Stack Engineering Intern

Fulton, MD

- Engineered a full-stack remote laboratory control platform using Python/Flask REST APIs and Docker, enabling real-time device synchronization and concurrent hardware orchestration for **25+** Johns Hopkins University graduate researchers.
- Developed **7+** reusable frontend components using HTML, CSS, and JavaScript for live instrument control interfaces.
- Containerized application services using Docker, ensuring reproducible deployment across Linux server environments.

Cooperative Institute for Satellite Earth System Studies (CISESS)

Jun 2023 - Jan 2024

Data Science Intern

College Park, MD

- Built Python/Unix pipelines with NumPy/Pandas to process **2,000+ daily NOAA satellite files** into climatology datasets.
- Automated drought classification across **6 years of satellite data**, visualizing anomalies in Matplotlib and validating against U.S. Drought Monitor.

EXTRACURRICULARS

App Development Club

Sep 2025 - Present

Recruitment | Events Coordinator | Web Development Bootcamp Member

College Park, MD

- Coordinated technical and community events for a **300+** member developer organization, boosting club engagement by **15%**.
- Pitched club programs to **1,000+** undergraduate students across campus CS lectures and onboarded prospective members.

Product Space

Feb 2026 - Present

Fellow

College Park, MD

- Selected for Product Space Fellowship (~**10% acceptance**) focused on product development and Figma-based UX design.
- Created an interactive capstone project in Figma and conducted market + user research to optimize profit and scale.

SKILLS & CERTIFICATIONS

Languages: Python, Java, C, JavaScript, TypeScript, SQL, Bash, HTML/CSS

Libraries/Frameworks: React, FastAPI, Flask, PyTorch, scikit-learn

Developer Tools/Technologies: Git, Linux, Docker, Kubernetes, OpenShift, PostgreSQL, MongoDB, SQLite, REST APIs, RunAI

Certification: AWS Certified Cloud Practitioner | Amazon Web Services

Jul 2026 - Jul 2029